INCH-POUND

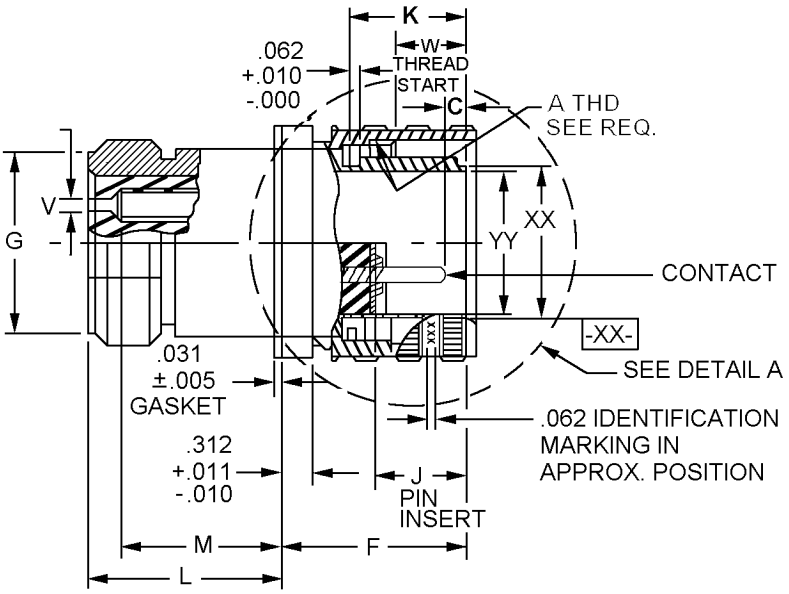
MS90558F
w/AMENDMENT 1
28 July 2010
SUPERSEDING
MS90558F
30 April 2009

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, WALL MOUNTING, (WITH COUPLING RING) CLASS L, (EQUIPMENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

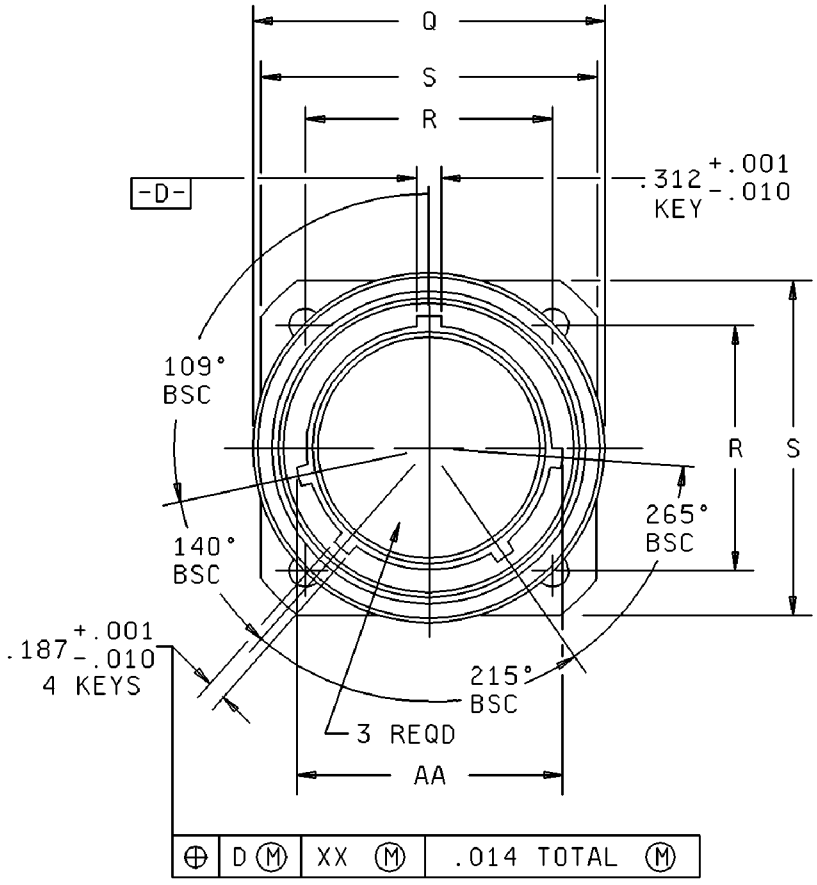
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22992.



Inches	mm
.005	0.127
.010	0.254
.011	0.279
.031	0.787
.062	1.57
.312	7.92

FIGURE 1. Plug.

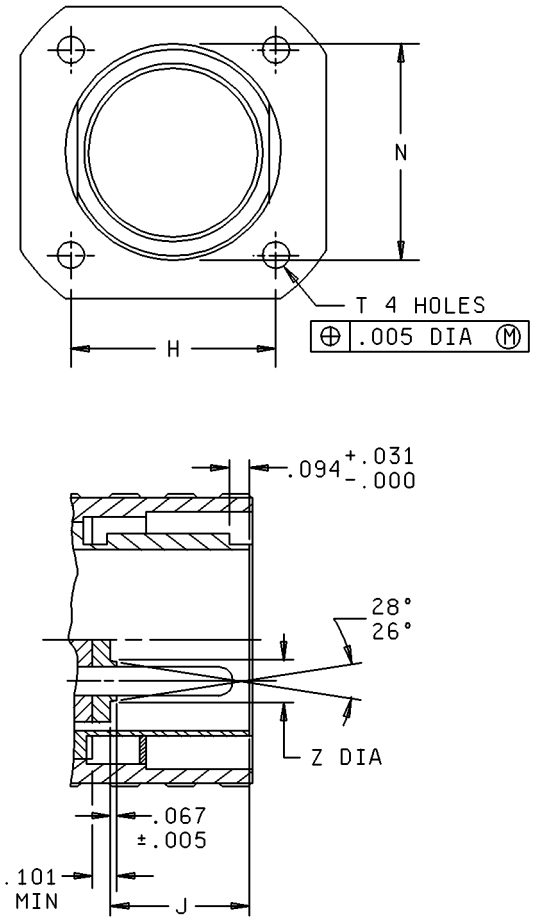
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Inches	mm
.312	7.92
.187	4.75
.010	0.254
.001	0.025

FIGURE 1. Plug - Continued.

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Inches	mm
.101	2.57
.094	2.39
.067	1.70
.031	0.79
.005	0.127

FIGURE 1. Plug - Continued.

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Contact size	Z dia	V dia wire sealing range
4 / 0	.614 (15.60) / .624 (15.85)	.700 (17.78) / .790 (20.07)
2 / 0	.520 (13.21) / .530 (13.46)	.530 (13.46) / .655 (16.64)
1 / 0-1	.471 (11.96) / .481 (12.22)	.457 (11.61) / .585 (14.86)
4	.339 (8.61) / .349 (8.86)	.356 (9.04) / .430 (10.92)
6	.292 (7.42) / .302 (7.67)	.300 (7.62) / .360 (9.14)

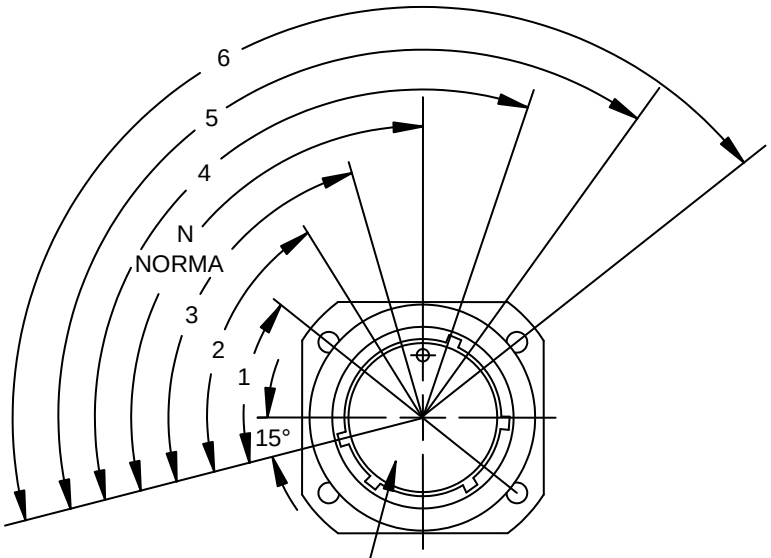
Shell size	C ± .047			F + .016 - .000	J ± .016	K ± .005	W thread start ± .031
	Phase	Neutral	Ground				
28	.196 (4.98)	.078 (1.98)	.078 (1.98)	2.639 (67.03)	1.370 (34.80)	1.557 (39.55)	1.062 (26.97)
32	.196 (4.98)	.078 (1.98)	.078 (1.98)	2.639 (67.03)	1.370 (34.80)	1.557 (39.55)	1.062 (26.97)
44	.196 (4.98)	.078 (1.98)	.547 (13.89)	2.998 (76.15)	1.589 (40.36)	1.776 (45.11)	1.281 (32.54)
48	.196 (4.98)	.078 (1.98)	.547 (13.89)	2.998 (76.15)	1.589 (40.36)	1.776 (45.11)	1.281 (32.54)
52	.196 (4.98)	.078 (1.98)	.547 (13.89)	2.998 (76.15)	1.589 (40.36)	1.776 (45.11)	1.281 (32.54)

Shell size	R (BSC)	S + .021 - .020	T ± .005	AA dia + .006 - .005	XX dia + .001 - .010	YY dia + .010 - .001
28	1.844 (46.84)	2.375 (60.33)	.177 (4.50)	1.672 (42.47)	1.522 (38.66)	1.370 (34.80)
32	2.062 (52.37)	2.625 (66.68)	.209 (5.31)	1.922 (48.82)	1.772 (45.01)	1.620 (41.15)
44	2.812 (71.42)	3.375 (85.73)	.281 (7.14)	2.672 (67.87)	2.522 (64.06)	2.296 (58.32)
48	3.000 (76.20)	3.625 (92.08)	.281 (7.14)	2.891 (73.43)	2.725 (69.22)	2.551 (64.80)
52	3.156 (80.16)	3.875 (98.43)	.281 (7.14)	3.141 (79.78)	2.975 (75.57)	2.801 (71.15)

FIGURE 1. Plug- Continued.

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Shell size	A Thread class 2B	G dia + .006 - .010	H + .001 - .010	L ± .031	M ± .047	N + .011 - .020	Q dia max
28	2.000-.1428P-.2857L	1.938 (49.23)	1.812 (46.02)	.959 (24.36)	.328 (8.33)	2.000 (50.80)	2.312 (58.72)
32	2.250-.1428P-.2857L	2.188 (55.58)	2.062 (52.37)	.959 (24.36)	.328 (8.33)	2.250 (57.15)	2.562 (65.07)
44	3.000-.1428P-.2857L	3.062 (77.77)	2.812 (71.42)	1.021 (25.93)	.390 (9.91)	3.125 (79.38)	3.531 (89.69)
48	3.250-.1428P-.2857L	3.312 (84.12)	3.062 (77.77)	1.021 (25.93)	.390 (9.91)	3.375 (85.73)	3.781 (96.04)
52	3.500-.1428P.2857L	3.562 (90.47)	3.312 (84.12)	1.021 (25.93)	.390 (9.91)	3.625 (92.08)	4.016 (102.01)



ALTERNATE KEY POSITIONS

FIGURE 1. Plug - Continued.

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Non-grounding assemblies: finish N

Shell size	Shell main key positions
	2 wire
	28 V dc
28	N (105°)
32	N (105°)
44	N (105°)
48	N (105°)
52	N (105°)

Grounding assemblies: finish C

Shell size	Shell main key positions						
	60 Hz and 400 Hz (see note 1)						
	1 ∅			3 ∅			3 ∅ (see note 2)
	2 wire		3 wire	4 wire			3 wire
	120 V	240V	120/240 V	120/208 V	240/416 V	277/480 V	450/480 V
28	4 (120°)	5 (135°)	4 (120°)	4 (120°)	5 (135°)	6 (150°)	---------
32	4 (120°)	5 (135°)	4 (120°)	4 (120°)	5 (135°)	6 (150°)	---------
44	4 (120°)	---------	4 (120°)	4 (120°)	5 (135°)	6 (150°)	1 (60°)
48	--------	---------	4 (120°)	4 (120°)	5 (135°)	6 (150°)	---------
52	--------	---------	4 (120°)	4 (120°)	5 (135°)	6 (150°)	---------

NOTES:
1. Discrimination of 60 Hz and 400 Hz assemblies is accomplished by alternate positioning of inserts.
   See applicable insert drawing for keying.
2. For Navy ground support equipment use only. Arrangement 44-51, 44-52 and 44-56 are not applicable.
3. Shell main key positions 2 and 3 not used.
4. Dimensions are in inches. Metric equivalents are given for information only.
5. For protective cover, see MS90564.
6. F dimension applies only when the coupling nut is fully engaged.

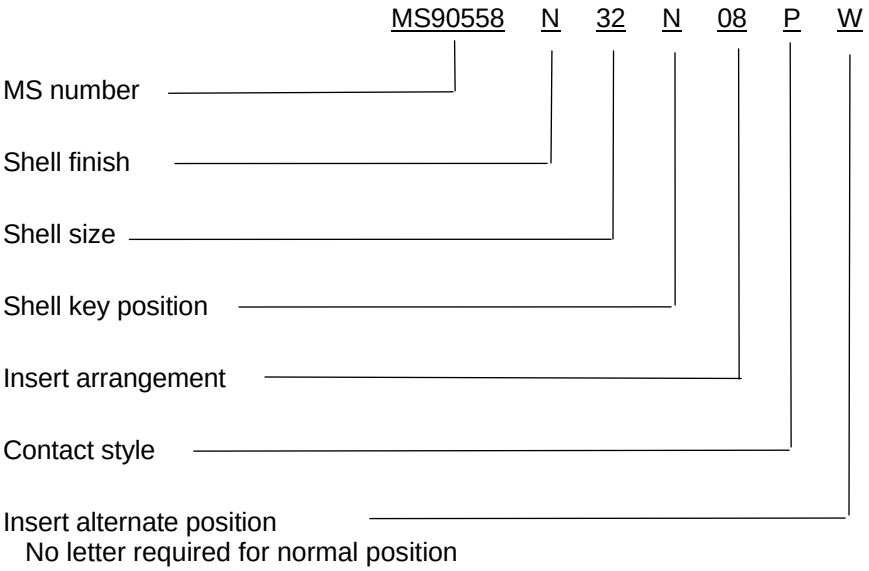
FIGURE 1. Plug - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.
Mates with receptacles shown on standard MS90557.
The Part or Identifying Number (PIN) shall include the size, finish and shell key alternate position.
Inserts and contacts are removable.
For final installation, contacts will be assembled in all insert holes.
Shell and nut finish shall be designated by the letter C (conductive) or N (non-conductive).
N designates normal key position. For alternate key positions see figure 1.
For insert arrangements (including alternate insert rotations W, X, Y and Z, where applicable), see MS90565, MS90567, MS14054, MS14055, and MS14057.
Covers specified in MS90564 shall be supplied with connector.
Pin contacts shall be in accordance with SAE- AS39029/48.

PIN example:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-22992, this document references the following:

MS14054
MS14055
MS14057
MS90557
MS90564
MS90565
MS90567
SAE-AS39029/48

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

Review activities:
Army – AR, AV, MI
Navy – AS, CG, MC, OS
Air Force – 19, 99

(Project 5935–2010-146)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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